

Quaternions

Quaternions can be described as a 4 dimensional number system which can be used for rotations.

$$i^2 = j^2 = k^2 = ijk = 1$$

A quaternion can be described as: $q = w + i + j + k$

Product of Imaginary numbers:

Product	i	j	K
K	-1	k	-j
J	-k	-1	i
I	j	-i	-1

Quaternion algebra:

Quaternions can be split into 2 parts a vector part $v = (x, y, z)$ and a scalar part ($s = w$)

$q = (s, v)$, with $s = w$ real and $v = (x, y, z)$ imaginary. A quaternion where $S = 0$ is called a pure quaternions and a quaternion with $V = 0$ is just a real number.

- The conjugate: $q^* = (s, -v)$
- Norm: $|q|^2 = |qq|$
- Inverse: $q^{-1} = \frac{q^*}{|q|^2}$

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Multiplying 2 Quaternions:

$$q_1 q_2 = (w_1 + x_1 i + y_1 j + z_1 k)(w_2 + x_2 i + y_2 j + z_2 k)$$

$$w_1(w_2 + x_2 i + y_2 j + z_2 k) = w_1 w_2 + w_1 x_2 i + w_1 y_2 j + w_1 z_2 k$$

$$x_1 i(w_2 + x_2 i + y_2 j + z_2 k) = x_1 w_2 i + x_1 x_2 i^2 + x_1 y_2 ij + x_1 z_2 ik = x_1 w_2 i - x_1 x_2 + x_1 y_2 k - x_1 z_2 j$$

$$y_1 j(w_2 + x_2 i + y_2 j + z_2 k) = y_1 w_2 j + y_1 x_2 ji + y_1 y_2 j^2 + y_1 z_2 jk = y_1 w_2 j - y_1 x_2 k - y_1 y_2 + y_1 z_2 i$$

$$z_1 k(w_2 + x_2 i + y_2 j + z_2 k) = z_1 w_2 k + z_1 x_2 ki + z_1 y_2 kj + z_1 z_2 k^2 = z_1 w_2 k + z_1 x_2 j - z_1 y_2 i - z_1 z_2$$

Collect the terms for (w,i,j,k)

$$w(w_1 w_2 - x_1 x_2 - y_1 y_2 - z_1 z_2)$$

$$i(w_1 x_2 + x_1 w_2 + y_1 z_2 - z_1 x_2)$$

$$j(w_1 y_2 - x_1 z_2 + y_1 w_2 + z_1 x_2)$$

$$k(w_1 z_2 + x_1 y_2 - y_1 x_2 + z_1 w_2)$$

$$q_1 q_2 = (w_1 w_2 - x_1 x_2 - y_1 y_2 - z_1 z_2) + i(w_1 x_2 + x_1 w_2 + y_1 z_2 - z_1 x_2) + j(w_1 y_2 - x_1 z_2 + y_1 w_2 + z_1 x_2) + k(w_1 z_2 + x_1 y_2 - y_1 x_2 + z_1 w_2)$$

Multiplying Quaternion by Pure quaternion

Pure(q) = W

$$qw = (w + xi + yj + zk)(\omega_x i + \omega_y j + \omega_z k)$$

$$w(\omega_x i + \omega_y j + \omega_z k) = w\omega_x i + w\omega_y j + w\omega_z k$$

$$xi(\omega_x i + \omega_y j + \omega_z k) = x\omega_x i^2 + x\omega_y ij + x\omega_z ik = -x\omega_x + x\omega_y k - x\omega_z j$$

$$yj(\omega_x i + \omega_y j + \omega_z k) = y\omega_x ji + y\omega_y j^2 + y\omega_z jk = -y\omega_x k - y\omega_y + y\omega_z i$$

$$zk(\omega_x i + \omega_y j + \omega_z k) = z\omega_x ki + z\omega_y kj + z\omega_z k^2 = z\omega_x j - z\omega_y i - z\omega_z$$

Collect the terms for (w,i,j,k)

$$-x\omega_x - y\omega_y - z\omega_z = (-x\omega_x - y\omega_y - z\omega_z)1$$

$$(w\omega_x i + y\omega_z i - z\omega_y i) = (w\omega_x + y\omega_z - z\omega_y)i$$

$$(w\omega_y j - x\omega_z j + z\omega_x j) = (w\omega_y - x\omega_z + z\omega_x)j$$

$$w\omega_z k + x\omega_y k - y\omega_x k = (w\omega_z + x\omega_y - y\omega_x)k$$

$$qw = (-x\omega_x - y\omega_y - z\omega_z) + (w\omega_x + y\omega_z - z\omega_y)i + (w\omega_y - x\omega_z + z\omega_x)j + (w\omega_z + x\omega_y - y\omega_x)k$$

Rotating vector using Quaternion

For rotation a pure quaternion is used to store the (x,y,z) vector in (i,j,k) axis, there's a set of very basic steps used to rotate a quaternion. Firstly the axis system is defined as (x,y,z | i,j,k) where x is roll, y is pitch and z is yaw these are all taken from body axis. The basic step to rotating a quaternion is to have a quaternion with a real part and a imaginary part to be rotated about for example rolling around x axis consist of using $q = \cos + i \sin$ but this is only per axis the formal description is:

$$q = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)(n_x i + n_y j + n_z k)$$

Term: $(n_x i + n_y j + n_z k)$

$(\sqrt{i^2 + j^2 + k^2} = 1)$ for $(n_x i + n_y j + n_z k)$ this specifically describes a vector pointing in a direction which rotates in the angle of θ . So say the vector points (1,0,0) $q = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)(1i + 0j + 0k) = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)i$ this defines the rotation around the body roll axis, the body orientation is defined by a pure quaternion $V = (i,j,k)$ where q isn't pure due to the cos term being real. Taking the quaternion q and multiplying by V allows the vector to be rotated about a axis. This is easy to keep track when applying a single rotation, meaning performing a rotation of roll, pitch or yaw at a time only require to:

Motion axis	q	q*
Roll	$q_x = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right)i$	$q_x^* = \cos\left(\frac{\theta}{2}\right) - \sin\left(\frac{\theta}{2}\right)i$
Pitch	$q_y = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right>j$	$q_y^* = \cos\left(\frac{\theta}{2}\right) - \sin\left(\frac{\theta}{2}\right>j$
Yaw	$q_z = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right>k$	$q_z^* = \cos\left(\frac{\theta}{2}\right) - \sin\left(\frac{\theta}{2}\right>k$

Example 1: given a Pure quaternion has vector $v = (1,0,0)$ to be rotated 90 degrees in the pitch axis j

$$\left(q_y = \cos\left(\frac{90}{2}\right) + \sin\left(\frac{90}{2}\right>j = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j\right) \quad (v = i + 0j + 0k = i) \quad (q_y^* = \cos\left(\frac{90}{2}\right) - \sin\left(\frac{90}{2}\right>j = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j)$$

$$q_y v = \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j\right)i = \frac{\sqrt{2}}{2}i + \frac{\sqrt{2}}{2}ji = \frac{\sqrt{2}}{2}i - \frac{\sqrt{2}}{2}k$$

$$q_y v q_y^* = \left(\frac{\sqrt{2}}{2}i - \frac{\sqrt{2}}{2}k\right)\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j\right) = \frac{1}{2}i - \frac{1}{2}ij - \frac{1}{2}k - \frac{1}{2}kj = \frac{1}{2}i - \frac{1}{2}k - \frac{1}{2}k - \frac{1}{2}i = -k$$

Example 2: the previous vector was rotated around $v = (1,0,0)$ to $v_2 = (0,0,-1)$ now roll the vector by 90 degrees using roll

$$\left(q_x = \cos\left(\frac{90}{2}\right) + \sin\left(\frac{90}{2}\right>i = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right) \quad (v = 0i + 0j - k = -k) \quad (q_x^* = \cos\left(\frac{90}{2}\right) - \sin\left(\frac{90}{2}\right>i = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i)$$

$$q_x v = \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right)(-k) = -\frac{\sqrt{2}}{2}k - \frac{\sqrt{2}}{2}ik = -\frac{\sqrt{2}}{2}k + \frac{\sqrt{2}}{2}j$$

$$q_x v q_x^* = \left(-\frac{\sqrt{2}}{2}k + \frac{\sqrt{2}}{2}j\right)\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i\right) = -\frac{\sqrt{2}}{2}k + \frac{1}{2}j + \frac{1}{2}j + \frac{1}{2}k = j$$

Example 3: take $v = (1,0,0)$ rotate it around yaw axis at 90 degrees around the Z axis

$$\left(q_z = \cos\left(\frac{90}{2}\right) + \sin\left(\frac{90}{2}\right>k = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}k\right) \quad (v = i + 0j + 0k = i) \quad (q_z^* = \cos\left(\frac{90}{2}\right) - \sin\left(\frac{90}{2}\right>k = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}k)$$

$$q_z v = \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}k\right)i = \frac{\sqrt{2}}{2}i + \frac{\sqrt{2}}{2}ik = \frac{\sqrt{2}}{2}i + \frac{\sqrt{2}}{2}j$$

$$q_z v q_z^* = \left(\frac{\sqrt{2}}{2}i + \frac{\sqrt{2}}{2}j\right)\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}k\right) = \frac{1}{2}i + \frac{1}{2}j + \frac{1}{2}j - \frac{1}{2}i = j$$

Observations of examples

Taking example 1,2,3 there's a few complications which arise, mainly being that the purpose of quaternions in rotation setting is that it can be used to rotate a body and be not be prone to gimbal lock, this means that the quaternion doesn't loose track of the axis since the result of example 2 and example 3 are the same yet should have different orientation shows a very important aspect on how the quaternions need to be setup for rotation this being the body and world axis system.

Rotating a body axis around world axis using Quaternions:

Since quaternion can be defined as a rotation it can also be used to define body orientation around a world axis, this is mainly what allows the Quaternion to be prone of gimbal lock its commutative meaning which translates what the next rotation will be based on the last this is a commutativity between the world and body. It can be set up as so:

Define the Quaternion:	Define the body axis:
Unit Quaternion: $Q = w + xi + yj + zk$	Body x (nose/roll axis) $\rightarrow i = (1,0,0)$
Start identity: $Q = w + 0i + 0j + 0k$	Body y (right/pitch axis) $\rightarrow j = (0,1,0)$
	Body z (up/Yaw axis) $\rightarrow k = (0,0,1)$

Example : pitch 90 degrees then roll 90 degrees

$$(q_y = \cos\left(\frac{90}{2}\right) + \sin\left(\frac{90}{2}\right)j = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j) \quad (v = i + 0j + 0k = i) \quad (q_y^* = \cos\left(\frac{90}{2}\right) - \sin\left(\frac{90}{2}\right)j = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j)$$

$$q_y i = \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j\right)i = \frac{\sqrt{2}}{2}i + \frac{\sqrt{2}}{2}ji = \frac{\sqrt{2}}{2}i - \frac{\sqrt{2}}{2}k$$

$$q_y i q_y^* = \left(\frac{\sqrt{2}}{2}i - \frac{\sqrt{2}}{2}k\right)\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j\right) = \frac{1}{2}i - \frac{1}{2}k - \frac{1}{2}k - \frac{1}{2}i = -k$$

$$q_y j = \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j\right)j = \frac{\sqrt{2}}{2}j + \frac{\sqrt{2}}{2}j^2 = \frac{\sqrt{2}}{2}j - \frac{\sqrt{2}}{2}$$

$$q_y j q_y^* = \left(\frac{\sqrt{2}}{2}j - \frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j\right) = -\frac{1}{2} + \frac{1}{2}j + \frac{1}{2}j + \frac{1}{2} = j$$

$$q_y k = \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j\right)k = \frac{\sqrt{2}}{2}k + \frac{\sqrt{2}}{2}jk = \frac{\sqrt{2}}{2}k + \frac{\sqrt{2}}{2}i$$

$$q_y k = \left(\frac{\sqrt{2}}{2}i + \frac{\sqrt{2}}{2}k\right)\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j\right) = \frac{1}{2}i - \frac{1}{2}k + \frac{1}{2}k + \frac{1}{2}i = i$$

This is a full pitch rotation of 90 degrees, now the body axis is defined by { nose = (0,0,-1), bottom (0,1,0), side tip (1,0,0) which makes this makes sense and keeps the orientation to a reference axis now to apply roll multiply it again by the roll quaternion but now with the new axis.

$$(q_x = \cos\left(\frac{90}{2}\right) + \sin\left(\frac{90}{2}\right)k = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}k) \quad (v = 0i + 0j - k = -k) \quad (q_x^* = \cos\left(\frac{90}{2}\right) - \sin\left(\frac{90}{2}\right)k = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}k)$$

$$[nose] q_x (-k) q_x^* = \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}k\right)(-k)\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}k\right) = \left(-\frac{\sqrt{2}}{2}k - \frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}k\right) = -\frac{1}{2}k + \frac{1}{2} - \frac{1}{2} - \frac{1}{2}k = -k$$

$$[wing tip] q_x j q_x^* = \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}k\right)j\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}k\right) = \left(\frac{\sqrt{2}}{2}j + \frac{\sqrt{2}}{2}i\right)\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}k\right) = \frac{1}{2}j + \frac{1}{2}i + \frac{1}{2}i - \frac{1}{2}j = i$$

$$[top] q_x i q_x^* = \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}k\right)i\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}k\right) = \left(\frac{\sqrt{2}}{2}i - \frac{\sqrt{2}}{2}j\right)\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}k\right) = \frac{1}{2}i - \frac{1}{2}j - \frac{1}{2}j - \frac{1}{2}i = -j$$

From this example there's a track of orientation per rotation initially it starts as 3 points [vi(1,0,0), vj(0,1,0), vz(0,0,1)] [vi=nose, vj = left wing, vz= top] so when all are 1 then they're just vectors perpendicular to each other the first step was to pitch 90 degrees (positive angle pitches down) this leaves vector vi = (0,0,-1) meaning nose pointing directly down, vj = (0,1,0) this direction doesn't change since it's the axis being rotated about and vz = (1,0,0) where now the top points where the nose was previously pointing which is the correct output for the given rotation. The second step is to roll the new orientation which is rolling a aircraft pointing up [vi(0,0,-1), vj(0,1,0), vz(1,0,0)] so when rolling another 90 degrees and applying the same pattern it yields [vi(0,0,-1), vj(1,0,0), vz(0,-1,0)] its all very logical but computationally expensive it serves well as a example but for application best to use quaternion composition.

Quaternion composition

Given a quaternion $Q = 1$, can be modelled as 3 vectors already $[v_i(1,0,0), v_j(0,1,0), v_z(0,0,1)]$ this can be used to describe orientation and keep its state without needing to decompose the vectors.